

Assignment day-5

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Task-1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

→ * motherboard :-

The main circuit board that connects all the hardware parts together so they can communicate.

* CPU (Central Processing Unit) :-

The "brain" of the PC that performs all calculations and executes program instructions.

* CPU cooler :-

A fan heatsink or liquid cooling block mounted on the CPU to prevent it from overheating.

* RAM (Random Access Memory) :-

The temporary short-term memory that holds active programs for quick access while the PC is on.

* Storage drive (SSD or HDD) :-

The permanent storage where your operating system, games and personal files are saved.

* GPU / Graphics card :-

A dedicated processor designed to handle rendering visual graphics, video output and 3D games.

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* PSU (Power supply unit) :-

converts electricity from your wall outlet into usable power and distributes it to each component.

* Case Fan :-

Internal fans that pull cool air into the cabinet and push hot air out to maintain good airflow.

Day 5.

(2) Take a clear photo or draw labeled diagram of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

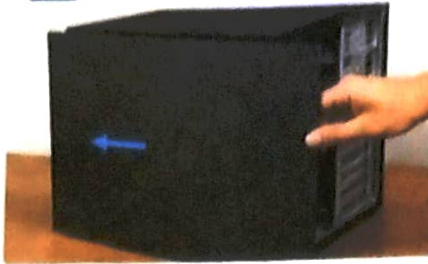
DESKTOP PC DISASSEMBLY – STEP BY STEP

1 Unplug All Cables



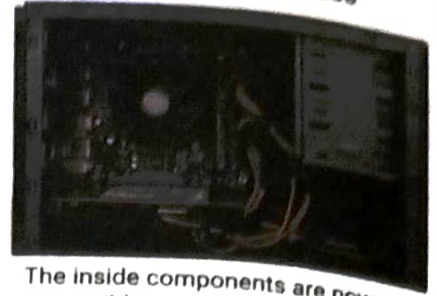
Turn off the PC and unplug the power cable and all other connected cables

2 Remove Side Panel



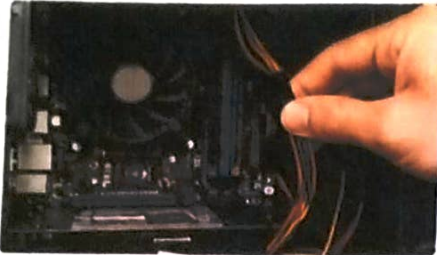
Remove the screws at the back and slide off the side panel to open the case

3 View Inside the Case



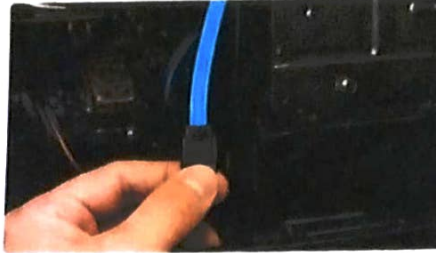
The inside components are now accessible.

4 Disconnect Power Cables



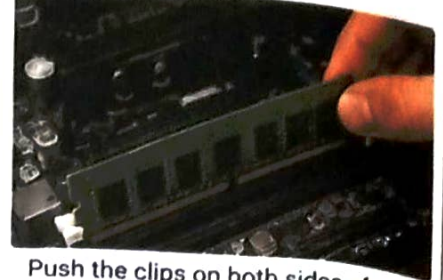
Disconnect the main power connector and other power cables from components.

5 Disconnect Data Cables



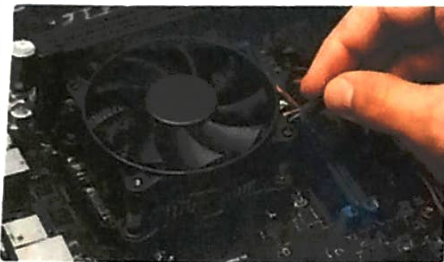
Disconnect SATA data cables from HDD/SSD and other components.

6 Remove RAM



Push the clips on both sides of the RAM stick and lift it out carefully.

7 Remove CPU Cooler



Unplug the cooler fan cable and unscrew or unclip the cooler. Then lift it off the CPU.

8 Remove HDD / SSD



Remove screws holding the HDD/SSD and take it out of the drive bay.

9 Remove Expansion Cards



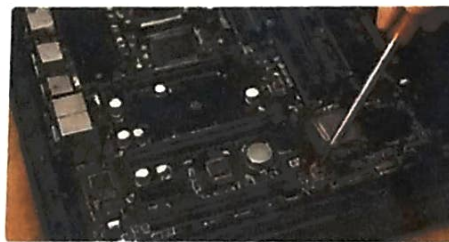
Unscrew the expansion card bracket and gently pull the card out of the slot.

10 Remove Power Supply



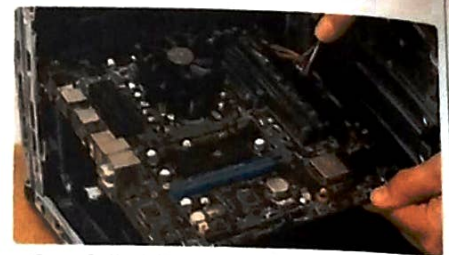
Unscrew the screws at the back of the PSU and slide it out of the case.

11 Remove Motherboard Screws



Remove all screws holding the motherboard to the case.

12 Remove Motherboard



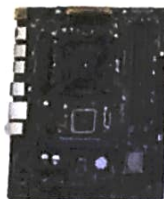
Carefully lift the motherboard out. The case is now empty.

13 Components Disassembled



All major components have been removed successfully.

DISASSEMBLED COMPONENTS



Motherboard



CPU Cooler



RAM



HDD / SSD



Graphics Card



Power Supply

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

→ When assembling a PC, order matters to avoid working in tight spaces or damaging sensitive components. Installing smaller, delicate parts onto the motherboard before mounting it into the cabinet saves time and prevents headaches.

1. Prepare the motherboard (CPU, RAM, M.2 SSD) :-
Do this outside the case on top the motherboard box).

Install the CPU carefully into its socket first, then snap the RAM sticks into their slots, and screw in the M.2 SSD storage drive. Installing this tiny, delicate parts outside the cabinet gives you plenty of room to work without hitting case walls.

2. Attach the CPU cooler :

(Before putting motherboard into cabinet).

Apply thermal paste to the CPU and mount the CPU cooler directly onto it. Cooler mounting brackets are much easier to align and secure when the motherboard isn't trapped inside a cramped metal case.

3. mount the Power supply unit (PSU) :-

(Place inside bottom of cabinet)

Screw the PSU into its designated slot at the bottom or back of the cabinet and feed the main power cables into the interior. Doing this before the motherboard keeps heavy cables from dragging across or damaging delicate motherboard circuits.

4. Install the motherboard into the cabinet :-

(Secure with standoffs and screws)

Align motherboard ports with the I/O shield on the ~~back~~ back of the case and screw it down onto the built-in metal standoffs. Standoffs are crucial because they prevent the metal case from touching the motherboard circuits and causing a short circuit.

5. Connect Front panel & motherboard cables :-
(Plug in power, USB & audio headers)

Plug the large 24-pin power cable, 8-pin CPU power cable and case cables (power button, USB ports, audio jacks) into the motherboard. Connecting these small, tedious cables now ensures you have clear hand room before a massive graphics card blocks your access.

6. Install the GPU (graphics card) :-
(Plugs into main PCIe slot)

Remove the expansion slot covers on the back of the case, push the GPU firmly into the top PCIe slot until it clicks, and screw it to the frame. The GPU goes to pin 1st because it is the largest, bulkiest component and would obstruct your hand during earlier cabling steps.

7. Plug in CPU power & Tidy cables :-
(Final connections before closing the case)

Connect the PCIe power cables from the PSU directly into the graphics card, double check that all cables are fully seated and secure loose wires with zip ties. Cable management ensures smooth internal airflow and keeps fans from catching on loose wires.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly / disassembly process, and explain why each is important.

→ Before opening up or disassembling a desktop PC, preparing the right tools and safety measures keeps both you and your expensive components safe from electrical damage or physical injury.

* Required Tools :-

* Phillips Head screwdrivers (#2 size) :-

The single most essential tool. Almost every screw inside a PC cabinet - from the motherboard standoffs to the CPU bracket - uses a standard #2 Phillips head.

* Zip ties or cable straps :-

Used to bundle loose wires together. Neat cable management keeps wires out of spinning fans and ensures smooth airflow inside the cabinet.

* Thermal paste (for CPU reassembly) :-

A heat-conducting gel applied between the processor and its cooler. It fills tiny microscopic air gaps to transfer heat effectively.

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- microfiber cloth & Isopropyl Alcohol (90%+) &

Used to clean off dry, old thermal paste from the CPU lid before re-applying a fresh layer.

- Small Trays or magnetic Bowl:-

Holds tiny chassis and mounting screws so they do not roll off your workspace or get lost inside the case.

- * Safety precautions :-

- Unplug the power cable completely :-

Never work on a PC while it's plugged into a wall outlet. Removing the main cord eliminates the risk of electrical shock.

- * Discharge Static Electricity (around yourself) :-

Touch a grounded metal object or wear an anti-static wrist strap before touching circuit boards. Static charges from your hands can fry delicate microchips without you even noticing.

- * Work on Hard, Flat, Non-conductive Surface :-

Assemble your PC on a wooden desk or dining table. Avoid working on carpets, rugs or beds, as soft fabrics build up static electricity and trap dust.

* Never Force Components :-

PC parts are designed to fit smoothly into their slots in only one orientation. If a stick of RAM or a graphics card isn't sliding in easily, stop and check the alignment rather than pushing hard and breaking the connector.

* Keep Liquid and Food Away :-

Ensure your hands are clean and dry, and keep drinks off the workspace. Spilling liquid onto exposed circuitry instantly ruins the hardware.